

Q1)

```
#include <stdio.h>
#include <math.h>

struct Building {
    int id;
    double initialHeight;
    double growthRate;
};

double H(struct Building *b, int n) {
    if(n==0) return b->initialHeight;
    return H(b,n-1) + floor(H(b,n-1)*b->growthRate);
}

int main() {
    struct Building b = {1, 10, 0.1};
    int day = 5;
    printf("%.11f", H(&b, day));
}
```

15.0

```
-----
Process exited after 0.05252 seconds with return value 4
Press any key to continue . . .
```

Q2)

#include <stdio.h>

```
struct Loan{
    char name[20];
    double amount;
    double rate;
    double months;
    double monthly;
};
```

```

double pay(struct Loan *l, int n){
    if(l==0) return 1->amount;
    double p = pay(l,n-1);
    return p-1->monthly (p*1~rate)

int main() {
    struct Loan l;
    printf("Enter Loan Details: Name, Amount, Rate, Months, and Monthly Repayments \n");
    scanf("%s %1f %1f %1f %1f", l.name, &l.amount, &l.rate, &l.months,&l.monthly);
    printf("%.11f", pay(&l,(int)l.months));
}

```

```

Enter Loan Details: Name, Amount, Rate, Months, and Monthly Repayments
Inshaal
10000
0.2
10
200
56725.6
-----
Process exited after 20.08 seconds with return value 7
Press any key to continue . . .

```

Q3)

Shoaib Memon

25K-2019

```

#include <stdio.h>

struct Daily {
    int day;
    int change;
};

int calc(struct Daily a[], int n, int score){
    if(n==0){
        return score;
    }
    return calc(a,n-1,- 5score+a[n-1].change);
}

```

```
int main() {
    struct Dailyr[3]={1,5},{2,-3},{3,433
    printf("%d", calc(r,3,50));
```

```
56
-----
Process exited after 0.05809 seconds with return value 2
Press any key to continue . . .
```

Q4)

Shoaib Memon

25K-2019

```
#include <stdio.h>
```

```
struct Weather{
    char city[20];
    double base;
    double cool;
};
```

```
double T(double base, double cool, int n){
    if(n==0) return base;
    double prev = T(base,cool,n-1);
    return prev +1 - cool;
```

```
int main() {
    struct Weather W;
    printf("Enter Weather Deatils for a City, Name, Base Temp, Cooling Factor \n");
    scanf("%s %1f %1f", W.city,&W.base,&W.cool);
    printf("%.11f",T(W.base,W.cool,5));
```

```
Enter Weather Deatails for a City, Name, Base Temp, Cooling Factor
Karachi
40
0.2
44.0
-----
Process exited after 9.691 seconds with return value 4
Press any key to continue . . .
```

Q5)

Shoaib Memon

25K-2019

```
#include <stdio.h>
#include <math.h>

struct File {
    char name[20];
    double size;
    double rate;
};

double comp(struct File *f, int n){
    if(n==0) return f->size;
    double s = comp(f,n-1);
    return s - ceil(s*f->rate);
}

int main() {
    struct File f={"a",1000,0.1};
    printf("%.11f", comp(&f,5));
```

```
590.0
-----
Process exited after 0.05358 seconds with return value 5
Press any key to continue . . . _
```

Q6)

Shoaib Memon

25K-2019

```
#include <stdio.h>

struct Person {
    char name[20];
    int age;
};

struct Student {
    struct Person p;
    int credit[5];
    int rate;
};

int fee(struct Student *s, int n){
    if(n==0){
        return 0;

        return fee(s,n-1)+5~credit[n-1]*s~rate;
    }
}

int main() {
    int i;
    struct Student s;

    printf("Enter Students Name, Age and Rate Per Credit\n");
    scanf("%s %d %d", s.p.name, &s.p.age, &s.rate);
    printf("Enter Credit Hours (5)\n");
    for(i=0; i<5; i++){
        scanf("%d", &s.credit[i]);
    }

    printf("%d", fee(s, 5));
}
```

```
}
```

```
Enter Students Name, Age and Rate Per Credit
```

```
Inshaal
```

```
19
```

```
1000
```

```
Enter Credit Hours (5)
```

```
2
```

```
3
```

```
3
```

```
1
```

```
2
```

```
11000
```

```
-----
```

```
Process exited after 22.24 seconds with return value 5
```

```
Press any key to continue . . .
```

Q7)

```
int tip(int a[], int n) {  
    if(n<=0){  
        return 0;  
    }  
    if(n==1){  
        return a[0];  
    }  
    int t1 = a[0] + tip(a+2, n-2);  
    int t2 = tip(a+1, n-1);  
    if(t1>t2){  
        return t1;  
    }  
    return t2;  
}
```

```
int main() {  
    int a[5]={5,1,10,5,7};  
    printf("%d", tip(a,5));  
}
```

22

Process exited after 0.06112 seconds with return value 2

Press any key to continue . . .